Israt Jahan Mou

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Professional Summary

Final-year Computer Science and Engineering undergraduate (CGPA 3.78/4.00) with hands-on experience in machine learning, deep learning, computer vision, explainable AI, and applied research. Built and deployed end-to-end ML systems for employee attrition prediction and IoT intrusion detection (model evaluation, imbalanced learning, explainability), and engineered PyTorch-based deep learning pipelines (YOLOv11, Video Swin Transformer) for IoT security and real-time surveillance threat detection. Seeking an AI/ML Engineering, Machine Learning, Deep Learning, or Computer Vision Internship.

Education

University of Asia Pacific (UAP), Dhaka, Bangladesh *Expected Graduation: 2027*
B.Sc. in Computer Science and Engineering — CGPA: 3.78/4.00

Technical Skills

Programming: Python, Java, SQL, JavaScript, HTML/CSS
Machine Learning: Scikit-learn, Linear Regression, Logistic Regression, K-Nearest Neighbors (KNN), Naive Bayes, Decision Trees, Random Forest, Support Vector Machines (SVM), Ensemble Learning, Classification, Regression, Clustering, Feature Engineering, Feature Scaling, Dimensionality Reduction, Imbalanced Learning, SMOTE, Cross-Validation, Model Evaluation, Hyperparameter Tuning, Threshold Tuning
Deep Learning: PyTorch, Neural Networks, Transfer Learning, Convolutional Neural Networks (CNNs), Transformer-Based Models
Computer Vision: YOLOv11, Video Swin Transformer, Object Detection, Spatio-Temporal Modeling
Explainable AI & Security: SHAP, Explainable Machine Learning, Adversarial Robustness (FGSM), Intrusion Detection
Data, Tools & Deployment: NumPy, Pandas, Matplotlib, Seaborn, Git, GitHub, Jupyter Notebook, Google Colab, Kaggle, VS Code, Joblib, Streamlit, Gradio, Hugging Face Spaces, Django, Django REST Framework, React, SQLite

Research and Selected AI/ML Projects

MOI-Lite + E-SATF: Lightweight Hierarchical IDS for IoT *Deep Learning, SHAP, TON-IoT, UNSW-NB15, CIC-IDS2017*

- Designed MOI-Lite, a lightweight two-stage IDS combining multi-scale dilated 1D-CNN blocks with lightweight self-attention for malicious IoT traffic classification; evaluated cross-dataset generalization and model compression across TON-IoT, UNSW-NB15, and CIC-IDS2017.
- Achieved 99.79% binary F1 and 94.37% multiclass macro F1; INT8 quantization reduced the model footprint to 65 KB for resource-constrained deployment.
- Developed E-SATF explanation-stability regularization for SHAP consistency; evaluated FGSM adversarial robustness, McNemar statistical comparisons, and inference latency. Manuscript in preparation for IEEE submission.

Multi-Stage Hybrid Architecture for Real-Time Surveillance Threat Detection *B.Sc. Final-Year Thesis, 2026*

- Engineered a multi-stage deep learning pipeline combining YOLOv11 (weapon/object detection) with Video Swin Transformer (spatio-temporal abnormal-behavior recognition) for real-time surveillance threat detection.
- Designed a risk-fusion mechanism combining spatial and behavioral signals into graded threat assessments, supporting real-time triage instead of binary alerts.

Employee Attrition Risk Predictor *Python, Scikit-learn, SMOTE, SHAP, Streamlit*

- Built and deployed an end-to-end ML system for employee attrition risk prediction on the IBM HR Analytics dataset (1,470 records) — preprocessing, imbalanced learning, model comparison, explainability, and deployment.
- Built a reproducible Scikit-learn pipeline (ColumnTransformer, OneHotEncoder, StandardScaler, SMOTE applied only to training data to prevent leakage); compared Logistic Regression, Random Forest, and Decision Tree via stratified 5-fold CV, selecting Logistic Regression for minority-class recall and stability.
- Optimized the classification threshold via precision-recall analysis and deployed a 30-feature Streamlit app with risk tiers, SHAP LinearExplainer-based explanations, and retention recommendations.

Additional Software Projects

Glam Girl — Full-Stack eCommerce Platform *Django, Django REST Framework, React, SQLite*

- Developed a full-stack eCommerce platform with a Django and Django REST Framework backend and React frontend, including product, cart, and order management.

AI Rasengan: Chakra Control System *JavaScript, MediaPipe Hands, HTML5 Canvas*

- Built a browser-based computer-vision interaction system using hand landmarks and gesture classification, with a real-time HTML5 Canvas particle engine.

Text Analyzer Pro & SQL Practice *Python, Tkinter, CLI, Pytest, MySQL*

- Developed a desktop/CLI text-analysis application supporting word-frequency statistics, search, JSON/TXT export, and automated testing. For SQL, applied relational database design, DDL/DML, constraints, joins, filtering, sorting, aggregation, and string functions.